

Local-Unitary Rigidity and Clifford Geometry of Six-Qudit AME Stabilizer Tensors

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Abstract

Let C be a linear $[6, 3, 4]_q$ MDS code and let $|\Psi_C\rangle = q^{-3/2} \sum_{c \in C} |c\rangle$ be its equal-phase Calderbank–Shor–Steane (CSS) state. We prove that every product-unitary intertwiner between two such states is local Clifford, for every prime power q . The proof shortens the stabilizer to four parties. Its nonidentity correlation tensor is diagonal on all $q^2 - 1$ local Weyl axes, and the rank-one contraction locus recovers those axes intrinsically. We apply this rigidity to an odd-prime-field non-Reed–Solomon pencil of six-point arcs. Projective, monomial-code, local-Clifford, and local-unitary equivalence are all classified there by one degree-eight quotient z . We also determine the split-torus versus $\mathrm{SL}_2(q)$ logical-Clifford phase. The same distinction between operator and scalar data governs the remaining results: fixed-copy scalar contractions are generically constant, but exact marginal and four-copy witnesses still detect special strata. A four-sheet transport operator explains one such rank-jump divisor, $(z - 2)(9z - 4)$.

1 Introduction

An absolutely maximally entangled state has maximally mixed reductions on every subsystem no larger than half the parties. At minimum support, a six-party state of local dimension q is naturally supplied by a length-six MDS code: if $C \leq \mathbb{F}_q^6$ is linear of dimension three, then

$$|\Psi_C\rangle = q^{-3/2} \sum_{c \in C} |c\rangle \quad (1.1)$$

is $\mathrm{AME}(6, q)$ exactly when C is $[6, 3, 4]_q$ MDS. This code–state dictionary is standard [RGRA18]. Our question is how much of the code geometry survives arbitrary single-party changes of basis.

The local-unitary versus local-Clifford problem is false for general stabilizer states [JCWY10]. It is therefore important that the following result concerns a specific MDS/CSS family and says more than equality of two orbit partitions.

Theorem 1.1 (Local-unitary rigidity). Let q be a prime power and let $C, D \leq \mathbb{F}_q^6$ be linear $[6, 3, 4]_q$ MDS codes. Suppose that, for a party permutation π , single-qudit unitaries $U_1, \dots, U_6$, and a phase θ ,

$$(U_1 \otimes \dots \otimes U_6) \pi |\Psi_C\rangle = e^{i\theta} |\Psi_D\rangle.$$

Then every U_i normalizes the finite-field Weyl system. In particular, the displayed equivalence is local Clifford.

The mechanism is visible in one reduced operator. Shortening both C and $C^\perp$ to four parties gives a q^2 -element stabilizer subgroup. At each retained party its $q^2 - 1$ nonidentity elements run bijectively through the nonidentity Weyl operators. After the identity term is

removed, the four-party reduced state is therefore a diagonal four-way tensor on the full Weyl basis. A contraction of such a tensor has rank one exactly on its coordinate axes. Those axes are intrinsic, so any product unitary carrying one reduced state to another must permute the local Weyl axes. This is precisely the Clifford condition.

Rains used the corresponding three-Pauli tensor to determine the automorphisms of qubit distance-two codes [Rai99, Theorem 13], and Van den Nest, Dehaene, and De Moor used that rigidity in their minimal-support criterion for qubit stabilizer states [VdNDDM05]. Our argument is a finite-field extension of this axis-recovery mechanism. To our knowledge, it gives the first LU-to-LC theorem in the exact scope of equal-phase CSS states from linear $[6, 3, 4]_q$ MDS codes, uniformly over prime-power q . The qualifier is essential: arbitrary phased minimal-support $\text{AME}(6, d)$ states form infinitely many LU classes [RL25], and neither non-linear orthogonal arrays nor general six-qudit stabilizer states are covered here.

Our main application is the ordered pencil

$$H_t = ((0, 1, 1 - t), (0, 1, t - 1), (1, 1 - t, 0), (1, t - 1, 0), (1, 0, -t), (1, 0, t)). \quad (1.2)$$

On its admitted odd-prime-field non-GRS locus, define

$$\begin{aligned} A(t) &= -4t(t - 1)^2, \\ B(t) &= (t^2 - t + 1)(t^2 - 3t + 1), \\ z(t) &= (B(t)/A(t))^2. \end{aligned} \quad (1.3)$$

Exact bracket identities show that z is the degree-eight projective quotient of the pencil. Theorem 1.1 upgrades its local-Clifford classification to the following complete local result.

Corollary 1.2 (Classification of the admitted pencil). For admitted parameters t, u over a prime field $\mathbb{F}_q$ with q odd, the following are equivalent:

$$H_t \sim_{\text{proj}} H_u \iff \ker H_t \sim_{\text{mon}} \ker H_u \iff \Psi_t \sim_{\text{LC}} \Psi_u \iff \Psi_t \sim_{\text{LU}} \Psi_u \iff z(t) = z(u),$$

where party permutations are allowed.

The rest of the paper separates classification from explicit scalar certificates. Section 2 fixes the arc, code, Weyl, and AME conventions. Section 3 proves the headline theorem. Section 4 proves the z -classification, and Section 5 determines the logical-Clifford phase: $\text{SL}_2(q)$ on the GRS locus and a split torus off it. Section 6 gives a uniform H3/GRS marginal separator, an exact four-copy $q = 13$ witness, and the generic-constancy obstruction for fixed-copy scalars. Section 7 derives the exceptional divisor of the four-copy witness from a nine-dimensional transport operator. Section 8 records the computational trust boundary and the precise scope of all results.

2 Six-arcs, MDS codes, and CSS states

Let $q = p^e$ and fix the additive character $\chi(a) = \exp(2\pi i \text{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(a)/p)$. On the standard basis of $\mathbb{C}^q$, indexed by $\mathbb{F}_q$, put

$$X(a)|x\rangle = |x + a\rangle, \quad Z(b)|x\rangle = \chi(bx)|x\rangle.$$

For $v = (a, b) \in \mathbb{F}_q^2$, write $W_v = X(a)Z(b)$, with any fixed consistent phase convention when only its projective axis is used. For the CSS stabilizer we use the exact lift

$$W_{(c, h)} = X(c)Z(h) = \bigotimes_{i=1}^6 X(c_i)Z(h_i), \quad (c, h) \in C \oplus C^\perp.$$

Since $h \cdot c' = 0$ for $h \in C^\perp$ and $c' \in C$, these lifts multiply without a residual phase and fix $|\Psi_C\rangle$. The q^2 single-party operators W_v form an orthogonal basis of the single-party Hilbert–Schmidt space. A unitary is Clifford when conjugation permutes their projective axes $\mathbb{C}W_v$.

A six-arc is an ordered sextuple of points of $\mathbb{P}^2(\mathbb{F}_q)$ with no three collinear. Choose nonzero column representatives and collect them in a 3-by-6 matrix H . The arc condition says that every three-column minor of H is nonzero. Consequently

$$C = \ker H \leq \mathbb{F}_q^6$$

is an $[6, 3, 4]_q$ MDS code, and its dual $C^\perp = \text{rowspace } H$ is MDS with the same parameters. Conversely, a parity-check matrix of such a code supplies a six-arc. Projectivities, independent rescaling of columns, and permutation of columns correspond exactly to monomial equivalences of the kernels.

Proposition 2.1 (Arc–AME dictionary). For a six-arc with kernel C , the state $|\Psi_C\rangle$ in (1.1) is a minimal-support stabilizer $\text{AME}(6, q)$ state. Its projective Pauli-label Lagrangian is

$$L_C = (C \oplus 0) + (0 \oplus C^\perp) \leq \mathbb{F}_q^{12}.$$

Monomial code equivalence, together with a party permutation, induces local-Clifford equivalence of the corresponding states.

Proof. The operators $X(c)$, $c \in C$, fix the equal superposition. The operators $Z(h)$, $h \in C^\perp$, fix it because $\chi(h \cdot c) = 1$. The two subgroups commute, and their label space has dimension six and is Lagrangian.

For any three-coordinate set I , puncturing C to I is bijective: injectivity follows from distance four and both spaces have dimension three. In the partial trace of $|\Psi_C\rangle\langle\Psi_C|$, equality on the complementary three coordinates therefore forces equality of the codewords, while the projection to I runs once through $\mathbb{F}_q^3$. Hence every three-party reduction is $q^{-3}I$. Smaller reductions are maximally mixed by tracing again, so the state is $\text{AME}(6, q)$. Its support has the minimum possible size q^3 .

A column multiplier $x \mapsto ax$ is a single-party Clifford operation: it sends $X(b)$ to $X(ab)$ and $Z(b)$ to $Z(a^{-1}b)$. Projective row operations do not change the kernel, and column permutations permute parties. This proves the last assertion. $\square$

For a party set S , let

$$L_C(S) = \{\ell \in L_C : \text{supp}(\ell) \subseteq S\}.$$

The stabilizer partial-trace formula gives

$$\rho_S^C = q^{-|S|} \sum_{\ell \in L_C(S)} W_\ell. \quad (2.1)$$

This formula is the bridge used in both the rigidity proof and the marginal invariants below. A Pauli-labelled quantity is not by itself an LU invariant; the invariant statements will always be expressed either as covariance of the reduced operators or as basis-free tensor contractions.

The GRS boundary has a simple geometric meaning. A dimension-three length-six generalized Reed–Solomon code is represented, up to column multipliers, by six points on a nonsingular conic in $\mathbb{P}^2$. We use “GRS state” for the state of any such code. The non-GRS pencil in Section 4 consists of arc matrices H_t away from that conic locus and from the other factors needed to keep the columns distinct and the displayed coordinates defined.

3 Four-party Weyl tensors and LU rigidity

We first isolate the linear-algebra statement that recovers a diagonal tensor's coordinate axes. It is the full-Weyl counterpart of the three-Pauli argument in [Rai99, VdNDDM05].

Lemma 3.1 (Diagonal-tensor axes). Let $r \geq 3$, let $E_1, \dots, E_r$ be N -dimensional complex inner product spaces, and choose orthonormal bases $\{e_{ij} : 1 \leq j \leq N\}$. If

$$T = \sum_{j=1}^N \lambda_j e_{1j} \otimes \cdots \otimes e_{rj}, \quad \lambda_j \neq 0, \quad (3.1)$$

then the coordinate axes in every E_i are intrinsic to T . Consequently, a product of invertible linear maps carrying one tensor of the form (3.1) to another is monomial in the displayed bases.

Proof. Contract the first factor against $x = \sum_j x_j e_{1j}^*$. The result is

$$T_x = \sum_j \lambda_j x_j e_{2j} \otimes \cdots \otimes e_{rj}.$$

Flattening from E_2 to $E_3 \otimes \cdots \otimes E_r$ gives a matrix of rank $\#\{j : x_j \neq 0\}$. Thus T_x has rank one exactly when x lies on a coordinate axis. Product equivalence preserves this rank-one contraction locus, and hence permutes the axes in the first factor. Apply the same argument in every factor. $\square$

Fix a four-party set S . Shortening an $[6, 3, 4]$ MDS code on the two coordinates outside S leaves a one-dimensional code. Choose nonzero shortened words $c^S \in C$ and $h^S \in C^\perp$. Each has support exactly S , and

$$L_C(S) = \{(ac^S, bh^S) : a, b \in \mathbb{F}_q\}. \quad (3.2)$$

At a party $i \in S$, projection of this plane to the local Pauli labels is

$$(a, b) \mapsto (ac_i^S, bh_i^S),$$

which is invertible because both coordinates are nonzero. Thus the $q^2 - 1$ nonidentity labels in $L_C(S)$ run bijectively through the nonidentity Weyl labels at every party of S .

Proposition 3.2 (A four-party marginal fixes the local Weyl axes). If a product unitary on S carries ρ_S^C to ρ_S^D , then every one of its four local factors is Clifford.

Proof. Subtract the identity term from (2.1) and view the result as a tensor in the four traceless local Hilbert–Schmidt spaces, each of dimension $q^2 - 1$. Conjugation preserves these spaces. By (3.2), after indexing the nonzero label $v = (a, b)$, the tensor has the form

$$T_S^C = \sum_{v \in \mathbb{F}_q^2 \setminus \{0\}} \lambda_v W_{L_1 v} \otimes W_{L_2 v} \otimes W_{L_3 v} \otimes W_{L_4 v}, \quad (3.3)$$

where every L_i is invertible and every coefficient λ_v is nonzero. This is a diagonal tensor with $N = q^2 - 1$.

Local conjugation fixes the identity and acts unitarily on each Hilbert–Schmidt space. Lemma 3.1 therefore forces each local adjoint action to permute the axes $\mathbb{C}W_v$, $v \neq 0$. It also fixes the identity axis, so the underlying unitary normalizes the projective Weyl group. This is the single-qudit Clifford condition. $\square$

Proof of Theorem 1.1. Absorb the party permutation by relabelling D . Partial trace of the six-party equivalence gives a product-unitary equivalence $\rho_S^C \sim \rho_S^D$ for every four-set S . Given a party i , choose a four-set containing it and apply Proposition 3.2. Thus every U_i is Clifford. $\square$

Only two marginals are needed if their supports cover all six parties; for example $S = \{1, 2, 3, 4\}$ and $S' = \{1, 2, 5, 6\}$. More generally, the proof applies whenever every party belongs to a support of size at least three whose supported stabilizer subgroup has order q^2 and projects bijectively onto all nonzero local Weyl labels. We use only the $[6, 3, 4]$ MDS specialization.

4 The non-GRS pencil and its quotient

Let H_t be the ordered matrix in (1.2), over a finite field of odd order, and put

$$G(t) = t^4 - 4t^3 + 7t^2 - 4t + 1.$$

The admitted non-GRS condition is

$$t(t-1)(t^2-t+1)(t^2-3t+1)G(t) \neq 0. \quad (4.1)$$

Factoring the twenty 3-by-3 column minors gives the first four factors, which keep the arc and quotient coordinates nondegenerate. The determinant for a conic through the six columns is a unit times G , so $G = 0$ is exactly the GRS boundary. Write $C_t = \ker H_t$ and $\Psi_t = \Psi_{C_t}$, and let A, B, z be as in (1.3).

Theorem 4.1 (Projective and Clifford quotient). For admitted t, u , the following are equivalent:

$$H_t \sim_{\text{proj}} H_u, \quad C_t \sim_{\text{mon}} C_u, \quad z(t) = z(u),$$

allowing arbitrary party permutations. If q is prime, these are also equivalent to $\Psi_t \sim_{\text{LC}} \Psi_u$.

Proof. Set

$$y(t) = \frac{(t-1)^2}{t}.$$

Then $A = -4t^2y$, $B = t^2(y^2 - 1)$, and

$$z = \frac{(y - y^{-1})^2}{16}.$$

It follows that $z(t) = z(u)$ exactly when

$$y(u) \in \{y(t), -y(t), y(t)^{-1}, -y(t)^{-1}\}. \quad (4.2)$$

Each of the four relations is realized by a direct projectivity and a party permutation. In order, one may use

relation	projectivity
$y(u) = y(t)$	$\text{diag}(1, (1-u)/(1-t), u/t)$
$y(u) = -y(t)$	$\text{diag}(1, (1-u)/(1-t), -u/t)$
$y(u) = y(t)^{-1}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & (u-1)/t \\ 0 & -u/(1-t) & 0 \end{pmatrix}$
$y(u) = -y(t)^{-1}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & (1-u)/t \\ 0 & -u/(1-t) & 0 \end{pmatrix}.$

The corresponding permutations are respectively the identity, (56) , $(35)(46)$, and (3546) . All denominators and determinants are nonzero on the admitted locus.

Conversely, the ten signed products of complementary 3-by-3 brackets of H_t form the multiset

$$\{+A, +A, +A, -A, -A, -A, +B, +B, -B, -B\}. \quad (4.3)$$

Projectivity and independent column rescaling multiply all ten products by one common scalar; party permutation changes at most their common sign. The multiplicities three and two in (4.3) recover $(B/A)^2 = z$. Hence projective equivalence forces equality of z . The equivalence with monomial codes follows from Section 2, and projective equivalence supplies an LC equivalence.

For the remaining direction assume that q is prime. Then a local Clifford induces a block $F_i \in \mathrm{SL}_2(q)$ on each local Pauli-label plane. Put $K_T = L_{C_t}(T)$ for every four-set T . If $i, j \in T$, projection $p_i : K_T \rightarrow \mathbb{F}_q^2$ onto the Pauli-label plane at i is an isomorphism by (3.2). Define the transition

$$M_T(i, j) = p_j p_i^{-1} : (\mathbb{F}_q^2)_i \longrightarrow (\mathbb{F}_q^2)_j$$

and, for distinct four-sets T, U containing i, j , the holonomy

$$\mathrm{Hol}(T, U; i, j) = M_U(j, i) M_T(i, j). \quad (4.4)$$

Local symplectic blocks F_i send this endomorphism to $F_i \mathrm{Hol} F_i^{-1}$. Reversing T, U takes its inverse, and a party permutation only relabels the data. Thus the multiset of

$$r(M) = \mathrm{Tr}(M)^2 / \det M$$

over the $15 \cdot 6 \cdot 5 = 450$ choices is an LC and party-permutation invariant.

Direct substitution of the shortened generators into (4.4) gives the following rational-function multiset; the symbolic identities and a direct-Lagrangian reconstruction are independently replayed by the paper-local certificate described in Section 8:

$$\begin{array}{r|l} 4 & 90 \\ -1/z & 144 \\ (2z+1)^2/z^2 & 24 \\ \text{each root of } X^2 - 8X + (8 - 16z - 1/z) & 96. \end{array} \quad (4.5)$$

Here 96 is the multiplicity of each root before specialization; coincident values have their multiplicities added. The identity

$$4B(t)^2 + A(t)^2 = 4G(t)^2$$

shows that the excluded GRS locus is exactly $z = -1/4$. Away from it, the constant 4 bin is isolated and the two quadratic roots are distinct. The bin containing $-1/z$ can merge with neither, one, or both of the weight-24 and one weight-96 contributions, so its multiplicity is 144, 168, 240, or 264. No other bin has one of these multiplicities. Hence the multiset canonically recovers $-1/z$, including every modular collision, and LC equivalence forces equality of z . $\square$

The map $t \mapsto z$ has degree eight: the four transformations in (4.2), followed by the quadratic relation $t+t^{-1} = y+2$, account for its eight sheets. The signed refinement $w = B/A$, $z = w^2$, is not quantum data. Both even and odd explicit party permutations reverse w , and their projectivities lift to monomial local Cliffords.

Proof of Corollary 1.2. Theorem 4.1 gives all equivalences except LU. Every LC is an LU, while Theorem 1.1 turns every LU intertwiner between the states Ψ_t, Ψ_u into an LC intertwiner. $\square$

The prime-field hypothesis is necessary for this scalar classification. Over $\mathbb{F}_{p^e}$, the basis permutation $|x\rangle \mapsto |x^p\rangle$ is a local Clifford and sends Ψ_t to Ψ_{t^p} , while $z(t^p) = z(t)^p$ need not equal $z(t)$. Concretely, in $\mathbb{F}_{25} = \mathbb{F}_5[s]/(s^2 - 2)$, both s and $-s = s^5$ are admitted, but

$$z(s) = 3 + 3s, \quad z(-s) = 3 + 2s.$$

Thus extension-field LC/LU classification requires at least a Frobenius quotient and, for the full Weyl normalizer, potentially further $\mathbb{F}_5$ -symplectic identifications. No such classification is claimed here.

5 The logical-Clifford phase

Regard any one party of an $\text{AME}(6, q)$ stabilizer tensor as an input and the other five as outputs. This is a $[[5, 1, 3]]_q$ encoding isometry. The kernel of the party action on Clifford tensor symmetries is

$$\mathcal{K}_C = \{(F_1, \dots, F_6) \in \text{SL}_2(q)^6 : (\bigoplus_i F_i)L_C = L_C\}.$$

For an encoder view, the *fixed-party symplectic kernel* means the image of $\mathcal{K}_C$ on its input block.

Theorem 5.1 (Logical phase). For an equal-phase CSS state from a six-arc over an odd prime field, the fixed-party symplectic kernel is

$$\begin{cases} \text{SL}_2(q), & \text{if the arc is conic/GRS,} \\ T = \{\text{diag}(a, a^{-1}) : a \in \mathbb{F}_q^\times\}, & \text{if the arc is nonconic.} \end{cases} \quad (5.1)$$

Thus every encoder view of a GRS tensor realizes the full one-qudit logical Clifford group from fixed-party symmetries. Off the GRS locus, the fixed-party logical group is the split torus. Any party-moving isoduality that exchanges the CSS X - and Z -spaces acts by the nontrivial Weyl-group element and generates the normalizer of T together with T .

Proof. Every four-party support carries the two-dimensional shortened label plane (3.2), and projection to the Pauli plane of any retained party is an isomorphism. If F_i is fixed at one party, requiring $\bigoplus_i F_i$ to carry each shortened plane to itself therefore determines the other five blocks through overlapping four-sets. These conditions are also sufficient: the shortened planes span L_C , because the weight-four shortened words span both three-dimensional MDS codes C and $C^\perp$.

The common diagonal anchor $\text{diag}(a, a^{-1})$ preserves $C_X \oplus C_Z^\perp$, so T is always present. Write a propagated collection as

$$F_i = \begin{pmatrix} \alpha_i & \beta_i \\ \gamma_i & \delta_i \end{pmatrix}, \quad D_\alpha = \text{diag}(\alpha_1, \dots, \alpha_6), \quad \text{and similarly for } D_\beta, D_\gamma, D_\delta.$$

Preservation of $L_C = C_X \oplus C_Z^\perp$ is equivalent to

$$\begin{aligned} D_\alpha C &\subseteq C, & D_\gamma C &\subseteq C^\perp, \\ D_\beta C^\perp &\subseteq C, & D_\delta C^\perp &\subseteq C^\perp. \end{aligned} \quad (5.2)$$

Apply these inclusions to the unique shortened word on each four-set. On two four-sets meeting in three parties, comparison through the projection isomorphisms shows that an off-diagonal coefficient vanishes at one party if and only if it vanishes at all parties. Hence a nondiagonal propagated collection makes D_γ or D_β nonsingular, and (5.2), together with

equality of dimensions, gives $D_\gamma C = C^\perp$ or $D_\beta C^\perp = C$. Conversely, such a diagonal code–dual equivalence satisfies (5.2) for the two elementary unipotent anchors; together with T they generate $\mathrm{SL}_2(q)$.

It remains to identify when the diagonal equivalence $C \sim C^\perp$ occurs. A parity-check matrix for C and a kernel basis for it represent associated, or Gale-dual, ordered six-arcs. The classical self-association theorem for six points in $\mathbb{P}^2$ says that these two sextuples are projectively equivalent exactly when the six original columns lie on a conic [HMSV08, Section 2.3]. Equivalently, solving the multipliers in (5.2) leaves the determinant of the six quadratic evaluation vectors. Since the columns form an arc, the conic is nonsingular. On it, the standard GRS dual multipliers realize the code–dual equivalence and hence every anchor in $\mathrm{SL}_2(q)$; off it, the preceding implication excludes every nondiagonal anchor.

Finally, an isoduality interchanges the two summands C_X and $C_Z^\perp$, so its input block swaps the two Weyl axes. It conjugates a to a^{-1} in T , giving $N(T)$ whenever such a party-moving symmetry is present. $\square$

This is an operational distinction, not a classical automorphism count. Including logical Paulis, the projective logical Clifford group has order $q^2|\mathrm{SL}_2(q)|$ on the GRS locus. At $q = 11$, the two non-GRS pencil classes have party-moving isodualities, so their logical group is $q^2N(T)$; the GRS and non-GRS orders are 159720 and 2420, respectively.

The MDS property also makes minimum operator pushing uniform. For any input party and nonidentity Pauli label, each of the ten four-sets containing that party gives a unique stabilizer representative supported there. Removing the input leaves a weight-three output Pauli. Thus each input Pauli has exactly ten minimum-support pushes, although the symmetry orbits of those pushes can distinguish finer projective classes. This uses only the projection isomorphism for the shortened planes, not the finite group calculation.

Arithmetic symmetry jumps do not change the phase. The characteristic-17 tetrahedral specialization lies on the GRS boundary and therefore retains the full $\mathrm{SL}_2(17)$ kernel. The characteristic-31 A_5 specialization is non-GRS and therefore retains only the split torus. These are symmetry enhancements inside the two phases, not exceptions to (5.1).

6 Explicit local-unitary invariants

Theorem 1.1 settles equivalence without scalar invariants. Scalar witnesses remain useful because they are short, basis-free certificates and because their failures explain what the operator-valued marginal retains.

6.1 A marginal-moment separator

For a four-party set T , let ρ_T be the reduced state and embed it into all six parties as

$$A_T = \rho_T \otimes I_{T^c}.$$

For an unordered triple $\{T, U, V\}$ of distinct four-sets, the number $\mathrm{Tr}(A_T A_U A_V)$ is invariant under arbitrary product unitaries; party permutation merely permutes the 455 triples. By (2.1),

$$\mathrm{Tr}(A_T A_U A_V) = q^{-\mathrm{rank}(L_C(T) + L_C(U) + L_C(V))}. \quad (6.1)$$

Here and below, ranks of label spaces and matching systems are over $\mathbb{F}_q$. Indeed, the three stabilizer sums have total normalization q^{-12} . Only triples of labels with sum zero survive the full trace; isotropy of L_C removes multiplication phases, and the kernel of the addition map from the three two-planes has size q^{6-r} , where r is the $\mathbb{F}_q$ -rank in (6.1).

Index a four-set by its omitted pair, an edge of K_6 . The rank in (6.1) is four exactly when the three corresponding chord lines concur in both the six-arc $\mathcal{A}$ and its Gale dual $\mathcal{A}^*$, represented by any kernel basis of the arc matrix. To see this, split each shortened CSS plane into its one-dimensional $X(C)$ and $Z(C^\perp)$ parts. For three omitted edges, the three shortened X -lines span dimension two precisely when the corresponding chords of $\mathcal{A}^*$ concur; the Z -lines give the same criterion for $\mathcal{A}$. Each span has dimension at least two, so their sum has rank four exactly when both concurrences hold.

Sixty triples are stars and concur universally. Any other three-edge graph that contains two adjacent edges but is not a star has two chords meeting at their common arc point and a third chord missing it, since the columns form an arc. The only remaining graph type is a perfect matching. Hence

$$\#\{q^{-4} \text{ moments}\} = 60 + b(\mathcal{A}, \mathcal{A}^*), \quad (6.2)$$

where b counts perfect matchings concurrent in both configurations.

Theorem 6.1 (Uniform H3/GRS separation). Let $\tau^2 = \tau + 1$, and let $\mathfrak{p}$ be a prime ideal of $\mathbb{Z}[\tau]$ of odd residue characteristic for which the reduction of the integral H3 (icosahedral) sextuple

$$((0, 1, 1 - \tau), (0, 1, \tau - 1), (1, 1 - \tau, 0), (1, \tau - 1, 0), (1, 0, -\tau), (1, 0, \tau))$$

is a six-arc. Write $k = \mathbb{Z}[\tau]/\mathfrak{p}$. If the reduced arc is non-GRS, its equal-phase state is not LU-equivalent, even after party permutation, to any six-point GRS state over k .

Proof. Exact integral chord determinants give ten common concurrent matchings for the H3 arc and its Gale dual. The other five determinant norms are -64 and -4 , so the count is exactly ten in every residue reduction covered by the theorem. This is an exact fifteen-matching calculation in $\mathbb{Z}[\tau]$, not a field sample. The H3 moment multiset therefore contains 70 copies of q^{-4} .

For six points S on a conic, a concurrent perfect matching is the orbit pairing of a fixed-point-free involution in $\text{Stab}_{\text{PGL}_2(q)}(S)$. A projectivity fixing three points is the identity, so this group acts freely on ordered triples and its order divides 120. The finite-subgroup classification of PGL_2 , including the modular cases [Fab23], then leaves the following bounds. Cyclic, dihedral, A_4 , and S_4 subgroups have at most 1, 6, 3, 6 fixed-point-free involutions, respectively. In the faithful degree-six action of A_5 , every involution fixes two points. In characteristic three the remaining semi-elementary and subfield cases of order dividing 120 again have at most six. Thus, outside characteristic five, the number of fixed-point-free involutions is in

$$\{0, 1, 2, 3, 4, 6\}.$$

Thus a GRS state has at most 66 copies of q^{-4} . In characteristic five the only additional six-point action above this bound is $\text{PGL}_2(5)$ on $\mathbb{P}^1(\mathbb{F}_5)$, which has ten; the H3 reduction is itself GRS there and is excluded by the theorem's hypothesis. Equation (6.1) is LU-invariant, so the gap proves the claim. $\square$

The five nonconcurrent H3 matchings form a one-factorization of K_6 . The marginal incidence recovers its S_5 stabilizer but forgets the even A_5 orientation. This is necessary: full local Fourier transform sends Ψ_C to $\Psi_{C^\perp}$, and an integral isoduality returns the H3 code through an odd pentad permutation in every odd reduction. The orientation is an LC-forgettable CSS electric/magnetic marking, not a separate LU class.

6.2 The exact dimension-thirteen four-copy witness

For m ket copies and m bra copies, choose a permutation $\sigma_i \in S_m$ at each party and contract ket copy a with bra copy $\sigma_i(a)$. If G generates a three-dimensional code, let $M_\sigma(G)$ be the homogeneous matching system

$$(u_a G)_i = (v_{\sigma_i(a)} G)_i \quad (1 \leq i \leq 6, 1 \leq a \leq m).$$

Counting its solutions gives the normalized contraction

$$I_\sigma(\Psi_C) = q^{3m - \text{rank } M_\sigma(G)}. \quad (6.3)$$

Theorem 6.2 (Exact four-copy separator at $q = 13$). The two admitted $q = 13$ pencil classes with $z = 4$ and $z = 12$ are not local-unitary equivalent, even after arbitrary party permutation.

Proof. At $q = 13$, the admitted pencil has two classes in the same complete three-copy and marginal-moment bucket, with $z = 4$ and $z = 12$. Use the explicit four-copy pattern

$$\sigma = (1, (03)(12), (23), (02)(13), (01)(23), (021)).$$

For $\pi \in S_6$, put $(\pi\sigma)_i = \sigma_{\pi^{-1}(i)}$, and sum (6.3) over all party permutations:

$$J_\sigma(\Psi) = \sum_{\pi \in S_6} I_{\pi\sigma}(\Psi).$$

Over $\mathbb{F}_{13}$, the $z = 4$ class has rank 21 for all 720 terms. The $z = 12$ class has rank 20 for 192 terms and rank 21 for 528. Therefore

$$J_\sigma(z = 4) = 720 \cdot 13^{-9}, \quad J_\sigma(z = 12) = 3024 \cdot 13^{-9}. \quad (6.4)$$

This proves arbitrary-LU inequivalence directly. The complete two- and three-copy contraction sets do not separate this pair; (6.4) makes no claim that four copies are globally minimal. $\square$

6.3 Why fixed-copy scalars cannot recover the pencil coordinate

Proposition 6.3 (Generic constancy). Fix m , a field of cardinality q , an irreducible reduced parameter variety X , and a regular generator-matrix chart $G(x)$ of constant code dimension. Every degree- (m, m) permutation contraction of the associated equal-phase states is constant on a dense open subset of X . The same holds for party-orbit sums and linear combinations. In the stable range $q \geq m$, every scalar LU-invariant of bidegree (m, m) is generically constant. For a reducible family, the statement holds separately on each irreducible component.

Proof. Formula (6.3) depends only on the rank of a matrix whose entries are regular functions of the family parameter. A maximal nonzero minor shows that this rank is constant on a dense open set. There are only finitely many diagrams at fixed m , so their generic-rank opens have a common dense intersection because X is irreducible. Over the complex state space, the first fundamental theorem for unitary invariants says that local copy-permutation diagrams span the bidegree- (m, m) invariants; when the local Hilbert-space dimension q is at least m , there are no additional dimension relations among these diagrams. This proves the last statement. $\square$

The dense open here is geometric. It need not contain a rational point over a particular small finite field; the assertion about such points is obtained only after checking that the open has an $\mathbb{F}_q$ -point.

For the pencil viewed over its rational-function field, the generic four-copy contraction values therefore generate only the constant field, not $\mathbb{Q}(z)$. The witness (6.4) detects a rank-jump stratum. In contrast, the four-party operator tensor (3.3) retains the Weyl axes and forces Cliffordness. This is the precise distinction between scalar blindness and marginal covariant rigidity.

7 The four-copy transport operator

We now explain the rank jump behind (6.4). Fix the six copy permutations

$$\sigma = (1, (03)(12), (23), (02)(13), (01)(23), (021)).$$

Let V be the three-dimensional message space and let $g_i : V \rightarrow k$ be the six coordinate covectors. Form a bipartite four-sheeted cover with vertices L_a, R_b , $0 \leq a, b < 4$, and a color- i edge from L_a to $R_{\sigma_i(a)}$. Put V at each vertex and k at each edge, using $g_{p(i)}$ as both restriction maps for a party assignment $p \in S_6$. A global section is a tuple $(x_a, y_b) \in V^8$ satisfying

$$g_{p(i)}(x_a - y_{\sigma_i(a)}) = 0 \quad (7.1)$$

on all 24 edges. The three-dimensional constant-section space is always present. Gauging it away leaves the 24-by-21 matching matrix in (6.3).

Let $U = k^4/k(1, 1, 1, 1)$, and let ρ_i be the action of σ_i on U . For each p , the MDS property makes $g_{p(3)}, g_{p(4)}, g_{p(5)}$ a basis of V^* . Define Q_p by

$$g_{p(i)} = \sum_{k=0}^2 (Q_p)_{ik} g_{p(3+k)} \quad (0 \leq i < 3). \quad (7.2)$$

Thus Q_p is the left block obtained by systematizing the party-reordered generator matrix on its last three columns; no pivot can vanish because every three coordinate covectors are independent. For the identity assignment,

$$Q_1 = \begin{pmatrix} a & b & c \\ a & c & b \\ d & d & d \end{pmatrix}, \quad \begin{aligned} a &= -2(t-1)^2, & b &= -(t^2 - t + 1), \\ c &= -(t^2 - 3t + 1), & d &= -2(t-1). \end{aligned}$$

If $Y_k \in U$ are the three free copy vectors, then (7.1) reduces to the 9-by-9 block transport operator

$$T_p = ((Q_p)_{ik}(\rho_i - \rho_{3+k}))_{i,k=0}^2. \quad (7.3)$$

The dimension identity

$$21 - \text{rank } M_{\pi\sigma}(G(t)) = 9 - \text{rank } T_p \quad (7.4)$$

holds over the field k , where $p = \pi^{-1}$ in the convention of Section 6. It follows by quotienting the three-dimensional constant-section kernel and then eliminating the three free systematic coordinates. Both steps are invertible on the six-arc locus.

Theorem 7.1 (Transport divisor). For a parameter on the admitted odd non-GRS pencil, there exists a party assignment $p \in S_6$ for which the sheaf (7.1) has a nonconstant section if and only if

$$(B^2 - 2A^2)(9B^2 - 4A^2) = 0,$$

or equivalently

$$(z - 2)(9z - 4) = 0. \quad (7.5)$$

At a generic point above $z = 2$, exactly 96 party assignments have rank 20; at a generic point above $z = 4/9$, exactly 192 do. Every other assignment has rank 21.

Proof. For the identity systematic chart, the coefficients satisfy

$$a = b + c, \quad d^2 = -2a, \quad A = a(c - b), \quad B = bc.$$

Common copy relabelling, a change of basis in V , and rescaling a coordinate covector act on T_p by invertible row and column operations. Modulo these operations, the 720 party assignments split into boundary types and three nonboundary relative-cover types. Here the type records the six relative copy permutations $\rho_j^{-1}\rho_i$, equivalently the even-cycle pattern in each pair of colored matchings. Expanding a representative determinant by signed cycle covers gives

$$\begin{aligned} P_- &= AB(3B - 2A), \\ P_+ &= AB(3B + 2A), \\ P_2 &= A(B^2 - 2A^2). \end{aligned}$$

With integral bases the determinants are respectively

$$64d^3AB(3B - 2A), \quad 64d^3AB(3B + 2A), \quad 64d^3A(B^2 - 2A^2).$$

All omitted factors are units on the admitted odd pencil. Every assignment outside these three types has a determinant that is a unit there. The signed cycle-cover ledger, the exhaustion of the 720 assignments, and a fraction-free determinant path are recorded in the paper-local certificate. The two signed types lie above $w = B/A = \pm 2/3$, hence $z = 4/9$; the axial type lies above $z = 2$. Fraction-free elimination also gives a nonzero 8-minor on each component, so the generic kernel is exactly one-dimensional.

The row/column equivalences act on assignments from the left by copy symmetries and from the right by color stabilizers. The axial support is one double coset with groups $S_4 \times C_2$ and $D_8 \times C_2$ and intersection C_2^3 . Its size is

$$\frac{|S_4 \times C_2| |D_8 \times C_2|}{|C_2^3|} = \frac{48 \cdot 16}{8} = 96.$$

Each signed support is one double coset with groups S_4 and $S_4 \times C_2$ and intersection S_3 , hence has size

$$\frac{|S_4| |S_4 \times C_2|}{|S_3|} = \frac{24 \cdot 48}{6} = 192.$$

The signed types are distinguished by the sign of w , and the axial cycle pattern differs from both, so the three supports are disjoint. The orbit certificate supplies explicit representatives, stabilizers, intersections, and the complete support sets. Equation (7.4) turns the one-dimensional transport kernels into rank 20 of the matching matrix and completes the proof. $\square$

The exceptional arithmetic belongs to this detector, not to Theorem 1.1. The exact boundary is:

char k	effect on the transport divisor	(7.6)
2	outside the odd-field pencil	
3	both admitted detector components disappear	
5	$z = 4/9$ lies on the excluded GRS boundary; $z = 2$ remains	
7	the two components merge; the rank histogram is $20^{288}21^{432}$	
11, 13, 41	pullback ramification only; no further rank drop.	

At $t = -1$, $A = 16$, $B = 15$, and the nonboundary exceptional primes are already visible in

$$B^2 - 2A^2 = -7 \cdot 41, \quad 9B^2 - 4A^2 = 7 \cdot 11 \cdot 13.$$

In characteristic 3, the axial factor becomes G^2 and the signed factors lie on $t(t - 1) = 0$, so no admitted component remains. In characteristic 5, $9B^2 - 4A^2$ is a unit times G^2 , removing only the signed component. In characteristic 7, $9B^2 - 4A^2 = 2(B^2 - 2A^2)$; the disjoint 96- and 192-element supports merge, and exact 8-minors exclude a larger kernel. At 11, 13, 41 the indicated pullback component has a double root at $t = -1$, but the same minors keep the rank drop equal to one.

The divisor records where one scalar contraction changes rank. It is not an LU-orbit boundary and, by Proposition 6.3, cannot be promoted to a generic coordinate.

8 Verification, literature, and scope

The rigidity theorem, the arc-AME dictionary, and generic constancy are conceptual proofs in the text. The projective quotient uses displayed bracket identities and projectivities. Its transition-map invariance and collision argument are proved in Section 4, while the 450 exact rational functions are certificate-supported. Section 5 proves the propagation and Gale-fixed implications; its $q = 11, 17, 31$ group evaluations are certificate-supported examples, not inputs to the uniform theorem. Section 6 proves the marginal trace/rank and incidence bridges; the H3 determinants and $q = 13$ contractions are certificate-supported. Section 7 proves the systematic transport reduction; the cycle-cover determinants, assignment exhaustion, exact minors, and double-coset support sets are certificate-supported.

The reproducibility package is in `supplement/`. From that directory,

```
python3 verify.py           checks all fifteen byte counts and SHA-256 digests,
python3 verify.py --replay  regenerates all seven canonical JSON certificates in memory.
```

The full replay uses only the Python standard library, deterministic integer, rational, polynomial, and finite-field arithmetic, and canonical JSON serialization. It includes independent paths appropriate to each claim: CSS versus direct-Lagrangian shortening, symbolic versus finite projectivity, chord concurrency versus stabilizer ranks, quotient minors versus finite-field elimination, and cycle-cover versus fraction-free transport determinants. The exact files, hashes, domains, expected outputs, and negative scopes are recorded in `supplement/EVIDENCE.md` and `supplement/EVIDENCE-MANIFEST.json`.

The trusted boundary is Python integer arithmetic, the exact-arithmetic implementations in the generators, deterministic Gaussian elimination, and the mathematical dictionaries connecting arcs, codes, stabilizers, and tensor contractions. The certificates check the bounded enumerations and symbolic identities that they record. They do not prove the conceptual correspondence between the implemented objects and the paper definitions; that correspondence is supplied in the preceding sections.

The accompanying Lean 4 artifact uses toolchain `v4.32.0-rc1`. Its aggregate terminal `RelativeConicArcs.Gates.AMELUAggregate` checks the complete arc-MDS-CSS-AME dictionary, the algebraic pencil quotient, the finite marginal graph counts, the contraction normalization and party-orbit identities, and the transport polynomial and characteristic-seven identities in one environment. The geometric classification, marginal-trace, logical propagation, finite contraction, and transport orbit steps enter separate hypothesis structures; the paper constructs or certificate-supports them as specified above, but the Lean modules do not. Importing the terminal therefore does not make them unconditional formal theorems. In particular, the headline LU-to-LC rigidity theorem, the fixed-copy generic-constancy proposition,

the party-moving logical normalizer, and the determinant and double-coset inputs of the complete transport theorem are not formalized. The companion `AMELUAggregateAxioms` terminal separates the three native finite graph counts from the ordinary kernel-checked declarations. For each conditional structure, the statement-adequacy ledger records its fields individually. Some fields are constructed in the displayed paper proof, such as the four deck projectivities and the marginal trace/rank reduction; some are mathematical implications supported by a paper-local certificate, such as the H3 determinant count and the four-copy rank evaluations; and some combine a prose construction with a certificate replay, such as the transport determinant and double-coset inputs. These are hypotheses of formalized implications, not additional Lean theorems.

The LU-to-LC mechanism has direct qubit ancestry. Rains recovered the three Pauli axes of distance-two code projectors [Rai99]; Van den Nest, Dehaene, and De Moor used minimal stabilizer supports to force every local factor of an LU equivalence to be Clifford [VdNDDM05]. General minimal-support AME equivalence permits behavior outside our equal-phase CSS hypotheses [BR20, RL25]. Our qualified novelty claim is only the full $q^2 - 1$ -axis, prime-power-qudit extension and its uniform $[6, 3, 4]_q$ MDS/CSS specialization. It is not a new LU-LC mechanism in the abstract.

Finally, none of the results classifies nonlinear orthogonal arrays, arbitrary minimal-support AME tensors, non-MDS stabilizer states, or all stabilizer states over prime-power dimensions. We do not claim that four copies are globally minimal, that $z = 2, 4/9$ are orbit exceptions, or that a fixed copy degree separates all classes as q varies. The logical-group theorem and the quantum part of the pencil classification are restricted to odd prime fields. Over extension fields, Frobenius basis permutations are additional local Cliffords and the scalar z is only Frobenius-covariant. The H3 theorem is restricted to the stated odd residue-field reductions, and the detector exceptions in (7.6) do not alter the characteristic-uniform rigidity theorem.

The central distinction is therefore between covariant and scalar data. A four-party marginal retains the full configuration of local Weyl axes, and that configuration rigidifies every product-unitary intertwiner. Fixed-copy scalar contractions discard those axes and are generically constant, although their rank-jump strata still reveal exact divisors. For the admitted pencil, the covariant rigidity returns the classical degree-eight quotient z as the complete LU coordinate.

References

- [BR20] Adam Burchardt and Zahra Raissi. Stochastic local operations with classical communication of absolutely maximally entangled states. *Physical Review A*, 102(2):022413, 2020.
- [Fab23] Xander Faber. Finite p -irregular subgroups of $\mathrm{PGL}_2(k)$. *La Matematica*, 2(2):479–522, 2023.
- [HMSV08] Benjamin Howard, John Millson, Andrew Snowden, and Ravi Vakil. A description of the outer automorphism of s_6 , and the invariants of six points in projective space. *Journal of Combinatorial Theory, Series A*, 115(7):1296–1303, 2008.
- [JCWY10] Zhengfeng Ji, Jianxin Chen, Zhaohui Wei, and Mingsheng Ying. The LU-LC conjecture is false. *Quantum Information & Computation*, 10(1–2):97–108, 2010.

- [Rai99] Eric M. Rains. Quantum codes of minimum distance two. *IEEE Transactions on Information Theory*, 45(1):266–271, 1999.
- [RGRA18] Zahra Raissi, Christian Gogolin, Arnau Riera, and Antonio Acín. Optimal quantum error correcting codes from absolutely maximally entangled states. *Journal of Physics A: Mathematical and Theoretical*, 51(7):075301, 2018.
- [RL25] Nithin Ramadas and Arul Lakshminarayan. Local unitary equivalence of absolutely maximally entangled states constructed from orthogonal arrays. *Journal of Physics A: Mathematical and Theoretical*, 58(12):125301, 2025.
- [VdNDDM05] Maarten Van den Nest, Jeroen Dehaene, and Bart De Moor. Local unitary versus local Clifford equivalence of stabilizer states. *Physical Review A*, 71(6):062323, 2005.